A Convex Optimization Framework for Fair Resource Allocation in Settlers of Catan

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1 Introduction

1.1 Motivation

Settlers of Catan is a strategical board game where players compete for control of resources and land that are generated with different probabilities and production rates across the board. Early game decisions on board placements tend to strongly influence resource access throughout the remainder of the game. These often lead to some players being at an advantage. The motivation behind this project is to understand how such resource disparities can be analyzed and minimized through convex optimization. This allows us to build a model based on expected resource production and relaxed placement decisions in order to study the fairness and efficiency tradeoffs that arise from initial allocations and placements.

1.2 Previous Works

The research that has been done on *Settlers of Catan* has mainly targeted designing artificially intelligent agents with the goal of maximizing the probability of winning the game. There are three main approaches that have been taken to acheive this and they often rely on heuristics, simulation-based planning,valued-based algorithms, or search algorithms such as Monte Carlo Tree Search to evaluate the best action given the uncertainty of the board. Szita, Chaslot, and Spronck found that Monte Carlo Tree Search can be successfully adapted to *Settlers of Catan*, a multiplayer stochastic board game with hidden information and complex decision spaces [1].

Further research has been done with alternative approaches for modeling player decisions and thus evaluating game strategy. Guhe and Lascarides designed a framework for evaluating and comparing different game strategies in *Settlers of Catan* through a computational simulation system and strategic analysis [2]. Their research uncovers how various heuristic patterns can influence behavior and outcomes in terms of winning.

Outside of game AI, convex optimization is a widely used area of study to determine fairness, efficiency, and constrained resource allocation in engineering and economic systems. Within game AI the focus of research has always been on winning the game and less on the interactive models of the agents. As seen in class, Boyd and Vandenberghe provide a comprehensive framework for modeling optimization problems with convex objectives and constraints.This further enables a rigorous analysis of tradeoffs between efficiency and fairness in allocation problems, such as the problem we pose in this paper [3].

While prior research on artificial Catan agents has focused mainly on maximizing individual success (winning) with heuristic or known algorithmic models, our work instead investigates a fairness-oriented optimization principle. By relaxing the optimization from integer to continuous decision making we design a convex formulation that regulates inequality caused by agents designed to be greedy or self serving in expected resource production, and we shift the focus from the previously studied competitive advantage to the distribution of resources across all players.

1.3 Intended Contributions

Convex Modeling of Resource Allocation in Catan: We formulate a convex optimization model that captures expected resource production from settlement placements using continuous relaxations of discrete game decisions. This enables the placement problem to be expressed as a tractable convex program.

Dual Interpretation of Fairness Constraints: We analyze the Lagrangian dual of the optimization problem and interpret the dual variables as shadow prices for contested board locations and spatial constraints. These provide insight into which regions of the board are most valuable under fairness-oriented objectives.

Empirical Evaluation on Realistic Board Instances: We evaluate the proposed model on multiple simulated Catan board configurations. The optimized allocations are compared against heuristic and random placement strategies using metrics such as expected production, inequality measures, and game outcomes.

Tradeoff Analysis Between Efficiency and Fairness: We explore how different fairness regularization parameters affect total expected resource production. This illustrates the tradeoff between maximizing efficiency and maintaining balanced starting conditions among players.

1.4 Organization of the Paper

This paper is organized as follows. Section 1 describes task assignments. Section 2 introduces the problem setting and reviews prior work on *Settlers of Catan*, game AI, and convex optimization, while also summarizing the intended contributions of this project. Section 3 presents the formal statement of the problem, including the primal formulation, dual formulation, Lagrangian, and KKT conditions. Section 4 describes our approach, highlighting the design choices, novelty, and computational complexity of the proposed convex optimization framework. Section 5 presents the experimental setup, results, conclusion, and proposals for future work. Then references are listed in the final unnumbered section before the appendix. And finally, the appendix provides additional derivations, board-model details, and supporting heuristics omitted from the main text for clarity.

2 Statement of the Problem

We address the problem of allocating settlement placements on the Catan board to reduce inequality of expected resource production across players. Full notation is defined in Appendix A.

Let P be the number of players, $L = 54$ the number of settlement nodes, and $K = 5$ the number of resource types. The production matrix $A \in \mathbb{R}^{L\times K}$ has entries

$$A_{lk} = \sum_{h\in\mathcal{H}(l)} p_{n_h} \cdot \mathbb{1}[\text{tile } h \text{ produces resource } k], \tag{1}$$

where $p_{n_h} = (6-|n_h-7|)/36$ is the dice probability (Appendix B). The relaxed placement variable $x_{pl} \in [0,1]$ indicates whether player p occupies node l. The scalar production score of player p is $v_p = \sum_{l,k} x_{pl} A_{lk} w_k$, where $\mathbf{w} = \mathbf{1}$. The board-average production per non-empty node, $\bar{a}$, is defined in Appendix A.

2.1 Primal Formulation

The self-balance penalty for Convex is

$$\Phi_{\text{Convex}}(X) = \left\| \boldsymbol{\rho}_c - \frac{\mathbf{1}^\top \boldsymbol{\rho}_c}{K}\mathbf{1} \right\|_2^2, \qquad \boldsymbol{\rho}_c = \mathbf{x}_c A, \tag{2}$$

which penalises imbalance across resource types. The phase-adaptive self-floor fraction is

$$\phi = \begin{cases} 1.2 & |\mathcal{S}_c| + |\mathcal{C}_c| < 3 \quad \text{(early game)}, \\ 0.7 & \text{otherwise}, \end{cases} \tag{3}$$

where $\mathcal{S}_c$ and $\mathcal{C}_c$ are CONVEX's settlements and cities. The primal convex QP is

$$\underset{X,\, t^{\text{opp}},\, v^{\text{opp}}_{\max}}{\text{maximize}} \quad (1+\lambda)\, t^{\text{opp}} - \lambda\, v^{\text{opp}}_{\max} - \mu\, \Phi_{\text{CONVEX}}(X) \tag{4}$$

subject to

$$\begin{aligned}
&\textstyle\sum_l x_{pl} \le b_p && \forall p && \text{(placement budget)} && (5)\\
&t^{\text{opp}} \le v_p && \forall p \neq c && \text{(min opponent production)} && (6)\\
&v^{\text{opp}}_{\max} \ge v_p && \forall p \neq c && \text{(max opponent production)} && (7)\\
&v_c \ge \phi\, \bar{a}\, b_c && && \text{(self-production floor)} && (8)\\
&\textstyle\sum_p x_{pl} \le 1 && \forall l && \text{(node capacity)} && (9)\\
&\textstyle\sum_p x_{pl} + \sum_p x_{pm} \le 1 && \forall (l,m) \in \mathcal{E} && \text{(distance rule)} && (10)\\
&\textstyle\sum_p x_{pl} = 0 && \forall l \in \mathcal{O} && \text{(occupied nodes)} && (11)\\
&0 \le x_{pl} \le 1 && \forall p, l && \text{(box constraints)} && (12)
\end{aligned}$$

where b_p is the placement budget for player p (initial: $b_p = 2$; mid-game: $b_p = \max(0, 5 - |\mathcal{S}_p| - |\mathcal{C}_p|)$), $\mathcal{O}$ is the set of occupied nodes, and $\mathcal{E}$ the set of adjacent node pairs. The objective is linear in $t^{\text{opp}}, v^{\text{opp}}_{\max}$ and concave quadratic in X through $-\mu\Phi_{\text{CONVEX}}$; all constraints are linear. The problem is therefore a convex quadratic program (QP).

Note that CONVEX's own score v_c does *not* appear in the fairness objective terms: the QP is designed to compress the *opponent* production distribution while keeping CONVEX a credible board participant via constraint (8). The distance-rule constraint (10) approximates the Catan settlement adjacency rule; cities are handled via a separate heuristic (Section 3.1).

2.2 Dual Formulation

We derive the Lagrangian dual of the primal QP (4)–(12) including all multipliers for all (p, l) index pairs.

Lagrange multipliers.

- $\alpha_p \ge 0$: budget (5), $p = 1, \ldots, P$.
- $\gamma_p \ge 0$: min-opponent (6), $p \neq c$.
- $\delta_p \ge 0$: max-opponent (7), $p \neq c$.
- $\eta \ge 0$: self-floor (8).
- $\beta_l \ge 0$: capacity (9), $l = 1, \ldots, L$.
- $\sigma_{lm} \ge 0$: distance rule (10), $(l, m) \in \mathcal{E}$.
- ν^p_l: occupied-node equality (11) (unrestricted in sign), $l \in \mathcal{O}$, $p = 1, \ldots, P$.
- $\mu^+_{pl} \ge 0$: upper box $x_{pl} \le 1$ (12).
- $\mu^-_{pl} \ge 0$: lower box $x_{pl} \ge 0$ (12).

Lagrangian.

$$\begin{aligned}
\mathcal{L} = {} & (1+\lambda) t^{\text{opp}} - \lambda v^{\text{opp}}_{\max} - \mu \Phi_c(X) \\
& + \sum_p \alpha_p (b_p - \textstyle\sum_l x_{pl}) + \sum_l \beta_l \Big(1 - \sum_p x_{pl}\Big) + \sum_{(l,m)\in\mathcal{E}} \sigma_{lm} \Big(1 - \sum_p x_{pl} - \sum_p x_{pm}\Big) \\
& + \sum_{p \neq c} \gamma_p (v_p - t^{\text{opp}}) + \sum_{p \neq c} \delta_p (v^{\text{opp}}_{\max} - v_p) + \eta (v_c - \phi \bar{a} b_c) \\
& - \sum_{l \in \mathcal{O}} \sum_p \nu^p_l x_{pl} + \sum_{p,l} \mu^+_{pl} (1 - x_{pl}) - \sum_{p,l} \mu^-_{pl} x_{pl}.
\end{aligned}$$

Stationarity w.r.t. t^{opp} and $v_{\max}^{\text{opp}}$.

$$\frac{\partial \mathcal{L}}{\partial t^{\text{opp}}} = 0 \;\Rightarrow\; \textstyle\sum_{p \neq c} \gamma_p = 1 + \lambda, \qquad \frac{\partial \mathcal{L}}{\partial v_{\max}^{\text{opp}}} = 0 \;\Rightarrow\; \textstyle\sum_{p \neq c} \delta_p = \lambda. \tag{13}$$

Stationarity w.r.t. x_{pl}. Let $d_{lk} = A_{lk} - \frac{1}{K}\sum_{k'} A_{lk'}$ and define the gradient of the penalty as $[\nabla_X \Phi_c]_l = 2\sum_k (\boldsymbol{\rho}_{c,k} - \bar{\rho}_c) d_{lk}$ for $p = c$, and zero for $p \neq c$. Then for all p, l:

$$\begin{aligned} &-\mu\,[\nabla_X \Phi_c]_l \cdot \mathbb{1}[p = c] - \alpha_p - \beta_l - \sum_{m:(l,m)\in\mathcal{E}} \sigma_{lm} - \sum_{m:(m,l)\in\mathcal{E}} \sigma_{ml} \\ &\quad + (\gamma_p - \delta_p) \sum_k A_{lk} w_k \cdot \mathbb{1}[p \neq c] + \eta \sum_k A_{lk} w_k \cdot \mathbb{1}[p = c] - \nu_l^p \cdot \mathbb{1}[l \in \mathcal{O}] - \mu_{pl}^+ - \mu_{pl}^- = 0. \end{aligned} \tag{14}$$

Dual problem. The dual function is $g = \sup_{X, t^{\text{opp}}, v_{\max}^{\text{opp}}} \mathcal{L}$. The dual problem minimises g subject to the non-negativity of all multipliers and equations (13)–(14). Strong duality holds by Slater's condition: any interior point (e.g., uniform $x_{pl} = \varepsilon$ for small $\varepsilon > 0$) is strictly feasible for a non-trivially occupied board.

2.3 KKT Conditions

Because the problem is a convex QP, the KKT conditions are necessary and sufficient for global optimality. The complete conditions are in Appendix D. The complementary slackness conditions with the clearest economic interpretation are:

$$\alpha_p(b_p - \textstyle\sum_l x_{pl}) = 0 \;\forall p, \qquad \beta_l\Big(1 - \textstyle\sum_p x_{pl}\Big) = 0 \;\forall l, \tag{15}$$

$$\gamma_p(v_p - t^{\text{opp}}) = 0 \;\forall p \neq c, \qquad \delta_p(v_{\max}^{\text{opp}} - v_p) = 0 \;\forall p \neq c. \tag{16}$$

A player's budget multiplier α_p is positive only if its budget is fully used; a node's price β_l is positive only if it is fully occupied.

3 Approach

3.1 Design Choices and Agent Architecture

The agent solves one or more convex programs at each game tick, following the strict priority order shown in Figure 1. The key design decisions are summarised below; supporting derivations are in the appendix.

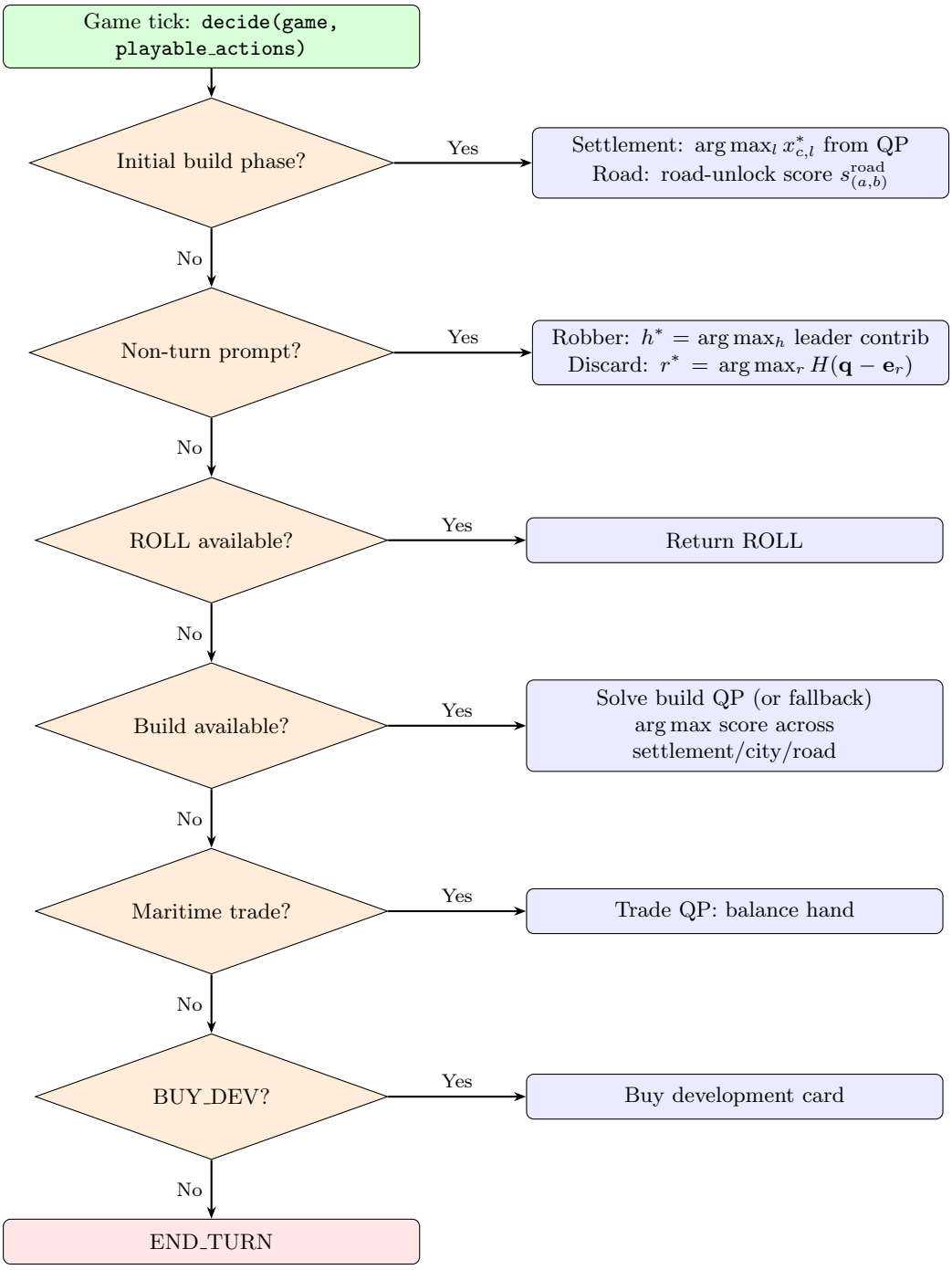


Figure 1: Decision flowchart of the ConvexAgent. At each game tick the agent follows a strict priority order, invoking the convex QP (or its closed-form fallback) for every build and trade decision.

Phase-adaptive self-floor. The floor fraction ϕ in constraint (8) switches between 1.2 (early game, $<$ 3 buildings) and 0.7 (mid-game). The higher early value forces Convex to compete for above-average nodes before dominant agents lock them up, preventing the self-sacrifice failure observed in the symmetric formulation.

Solver and fallback. The placement QP is solved via CVXPY with CLARABEL → SCS → ECOS fallback. When all solvers fail, the agent uses the closed-form dual-criterion score (Appendix E)

$$s_l = 0.6\, s_l^{\text{block}} + 0.4\, s_l^{\text{help}}, \tag{17}$$

which requires no solver and runs in $O(PL)$.

Build scoring. Settlements and roads use the LP node-score vector $x^*_{c,\cdot}$. Roads are scored by the maximum fairness score of unoccupied nodes reachable from either endpoint (road-unlock heuristic, Appendix F). City upgrades use an urgency-weighted score (Appendix G)

$$s_l^{\text{city}} = (\textstyle\sum_k A_{lk} w_k)\left(1 + \dfrac{\max(0,\, v_{\text{med}}^{\text{opp}} - v_c^{\text{cur}})}{v_{\text{med}}^{\text{opp}} + \varepsilon}\right). \tag{18}$$

Trade QP. Before building, the agent solves a hand-balancing QP (Appendix H):

$$\underset{\mathbf{t}^{\text{in}},\mathbf{t}^{\text{out}} \geq \mathbf{0}}{\text{maximize}}\ \mathbf{1}^\top \mathbf{q} - 0.5 \left\| \mathbf{q} - \tfrac{\mathbf{1}^\top \mathbf{q}}{K} \mathbf{1} \right\|_2^2, \quad \mathbf{q} = \mathbf{q}_0 + \mathbf{t}^{\text{in}} - \mathbf{t}^{\text{out}}, \tag{19}$$

subject to $\mathbf{q} \geq \mathbf{0}$ and the port-ratio constraint $\mathbf{1}^\top \mathbf{t}^{\text{in}} \leq \rho^{-1} \mathbf{1}^\top \mathbf{t}^{\text{out}}$, $\rho \in \{2, 3, 4\}$.

Robber and discard. The robber is placed on the hex maximally reducing the leader's expected production (28) (Appendix I). When forced to discard, the agent drops the resource maximising post-discard hand entropy (29) (Appendix J).

The full codebase is available at [10] and built on top of the `catanatron` simulator [7].

3.2 Novelty

Prior Catan AI uniformly targets win-rate maximisation via MCTS, alpha-beta search, or learned value functions [1, 2, 4, 5]. Our work differs in three ways.

Fairness as the primary objective. We are the first to formulate a Catan agent whose sole objective is reducing resource inequality. The QP minimises opponent production spread Δ^{opp} as a tractable linear surrogate for the Gini coefficient, producing a regulator that operates *within* the game economy.

Asymmetric opponent-aware formulation. Unlike symmetric welfare formulations (e.g., maximin [3]), our QP excludes CONVEX's own score from the fairness terms. Early symmetric designs caused CONVEX to self-sacrifice, paradoxically maximising inequality. The opponent-focused formulation with the phase-adaptive floor resolves this.

Integrated multi-decision QP pipeline. We apply convex optimisation at *every* decision type — placement, city upgrades, maritime trades, robber placement, and discarding — within a single per-turn pipeline with a guaranteed closed-form fallback. No prior Catan agent integrates convex programming across all decision points.

3.3 Computational Complexity

The placement QP has $PL + 2$ variables and $O(P + PL + L + |\mathcal{E}|)$ constraints. For $P = 3$, $L = 54$, $|\mathcal{E}| = 72$, this gives 164 variables and ≈ 300 constraints — solved in ≈ 4 ms on average (Table 3). The trade QP has $2K = 10$ variables. The fallback (17) runs in $O(PL)$ with no solver overhead, providing a constant-time safety net.

4 Results, Conclusions, and Future Work

4.1 Results

The initial formulation, as in the project outline, treated all players symmetrically in a single fair-allocation LP. The objective was to maximize a common floor t (with $v_p \geq t$ for every player p) and to penalize each

player's resource imbalance $\|\mu_p - \bar{\mu}_p \mathbf{1}\|_2^2$. The agent was just one of the p in this objective, with no separate guarantee on its own production. The robber stage chose the hex that most reduced the range of player scores. Trading was a per-turn convex problem to increase total resources and balance the hand. The design was intentionally "fairness-first": the same criterion applied to every player, including the agent.

Initial experiments with this symmetric formulation showed that the ConvexAgent often chose weak placements and lost more often than baseline agents. The LP was improving fairness in part by reducing the agent's own expected production, i.e., the agent was self-sacrificing. Metrics such as win rate and Gini made this clear: the convex agent lost more often and sometimes increased inequality because it left strong nodes to others. These results showed that the original formulation was not aligned with a competitive fairness objective. They motivated a reformulation that keeps opponent-focused fairness (raising the floor and compressing spread among opponents) while adding a phase-adaptive self-production floor and a self-imbalance penalty, so the agent no longer sacrifices its own position to improve the symmetric objective.

4.1.1 Experimental Setup

We report results from two complementary evaluations:

1. **Initial Placement Only (QP λ Sweep)**
   - We sample 500 random seeds (boards), with 3 players, and sweep $\lambda \in [0, 2]$ using 10 evenly spaced values (i.e. every ≈ 0.222). For each seed and each λ, we solve the initial-placement QP allocation and compute fairness and efficiency metrics from expected production. We also compare against baseline placement methods (Random, WeightedRandom, Greedy).
2. **Full-Game Simulation (Pairwise Lineup Evaluation)**
   - We evaluate all 35 distinct 3-agent lineups from the set {AB, CONVEX, GREEDY, MCTS, R, VALUE, WR}, running 500 games per lineup, for a total of 17,500 games. We summarize both competitive performance (win rate, final Victory Points (VP)) and fairness-style outcome diagnostics (VP Gini, VP range, resource-holdings entropy).

Experiment	**QP λ Sweep**	**Pairwise Full-Game**
Players per Game	3	3
Seeds/Boards	500	–
Lineups	–	35
Games per Lineup	–	500
Total Games	–	17,500
λ Grid	$[0, 2]$, 10 values ($\Delta \approx 0.222$)	fixed

Table 1: Summary of Experimental Configurations Used.

As summarized in Table 1, we separate (i) a QP-only initial placement study that measures fairness/efficiency on expected production from (ii) a full-game study that measures competitiveness and outcome parity over 17,500 simulated games.

Agents Used throughout the Experiments:

- AB (`AlphaBetaPlayer`): Performs an adversarial search (minimax) with alpha-beta pruning, attempting to choose actions that perform well under an opponent model by looking ahead in a game tree.
- CONVEX (`ConvexAgent`): Uses a convex optimization objective to regulate behavior, aiming to shape allocations and actions toward fairness or balance goals using "two-step" lookahead.
- GREEDY (`GreedyAgent`): Uses a myopic heuristic to evaluate available actions using an immediate reward proxy (such as VP gain, build potential, or immediate resource/value gain) and picks the best current scenario, without looking ahead.

- MCTS (`MCTSPlayer`): Uses Monte Carlo Tree Search by running simulations based off candidate actions and chooses actions with the best estimated long-run outcome.
- R (`RandomPlayer`): Chooses actions uniformly at random with no strategy.
- VALUE (`ValueFunctionPlayer`): Uses a learned value function to score states/actions, choosing actions that maximize predicted long-term value.
- WR (`CustomWeightedRandomPlayer`): Uses finetuned weights that typically favor building: cities over settlements, and settlements over development cards, rather than a pure random strategy.

4.1.2 Initial Placement Only: What Does λ Change in the QP Allocation?

Figure 2 shows how expected-production Gini, total expected production, and entropy vary with λ for the QP solution. Over $\lambda \in [0, 2]$, these curves are nearly flat for QP, indicating that within this tested range, the allocation is primarily shaped by feasibility constraints and the joint-allocation structure rather than the precise λ value.

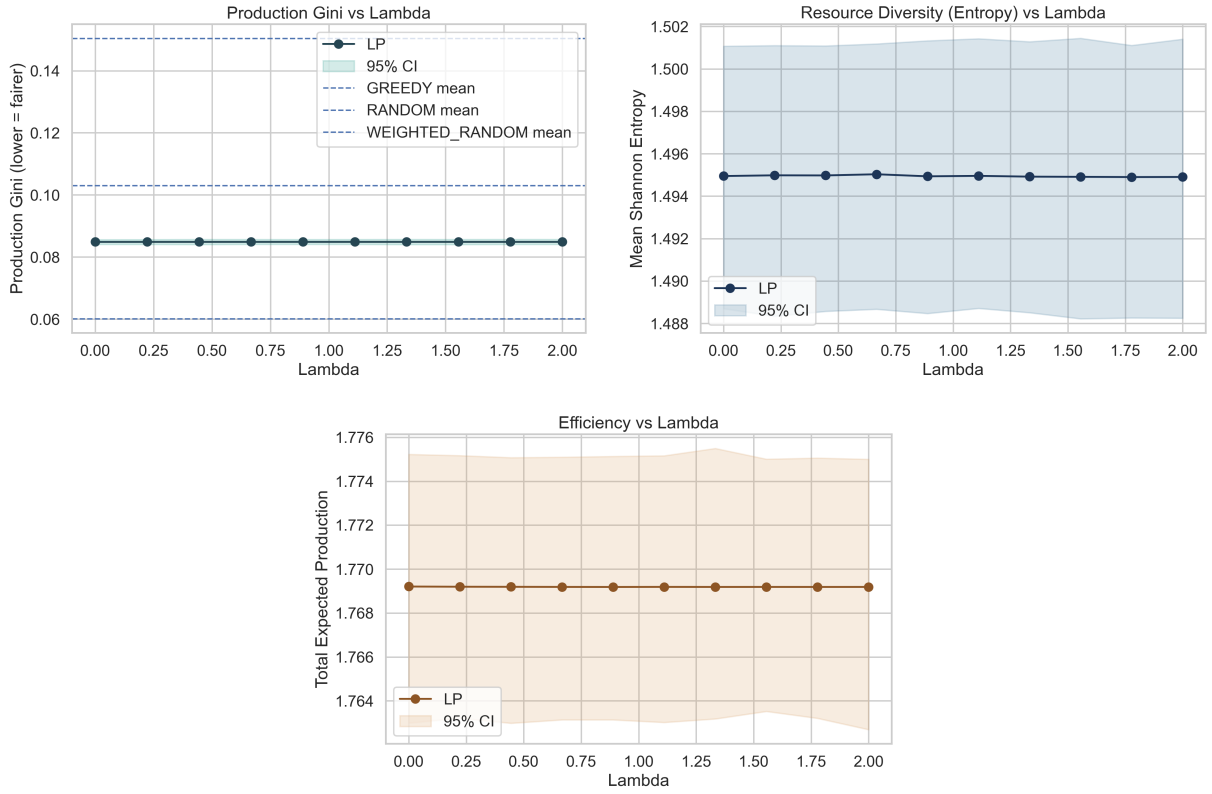


Figure 2: Initial-placement λ sweep (500 seeds, 3 players). For the QP method, expected-production Gini, mean entropy, and total expected production change only minimally over $\lambda \in [0, 2]$.

To make the cross-method differences concrete (beyond the near-flat λ trends), Table 2 reports a representative snapshot at the swept λ value closest to 0.5 ($\lambda \approx 0.444$).

Method	Gini (95% CI)	Total Prod. (95% CI)	Entropy (95% CI)	Max/Min (95% CI)
QP	0.085 [0.084,0.086]	1.77 [1.76,1.78]	1.495 [1.488,1.502]	1.515 [1.507,1.523]
GREEDY	0.060 [0.059,0.061]	2.12 [2.11,2.12]	1.214 [1.203,1.225]	1.315 [1.309,1.322]
WEIGHTED_RANDOM	0.103 [0.101,0.104]	1.83 [1.83,1.84]	1.077 [1.067,1.087]	1.697 [1.680,1.715]
RANDOM	0.150 [0.147,0.153]	1.62 [1.61,1.63]	0.929 [0.918,0.940]	2.321 [2.260,2.382]

Table 2: Initial-placement metrics at the swept λ value closest to 0.5 ($\lambda \approx 0.444$).

This snapshot aligns with Figure 3: GREEDY maximizes total expected production, while QP yields substantially higher entropy (more balanced resource profiles) at a moderate efficiency cost. While λ has limited effect for QP in this range, the *choice of method* has clear effects. Figure 3 compares fairness-efficiency tradeoffs and diversity across methods. GREEDY achieves the highest total expected production and low inequality (low Gini), but exhibits substantially lower entropy, indicating less balanced resource profiles. In contrast, QP yields the highest entropy among methods, at the cost of reduced total expected production relative to GREEDY. Both random baselines underperform across these metrics, with RANDOM exhibiting the lowest production and highest inequality.

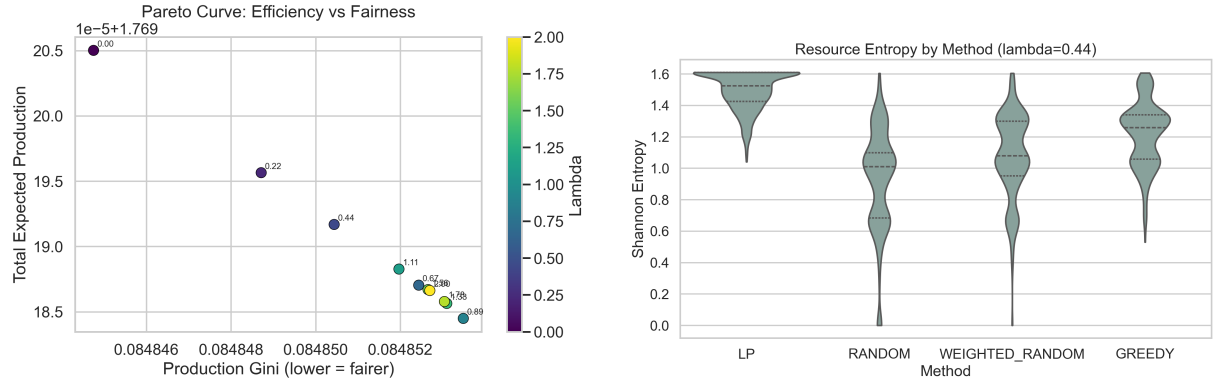


Figure 3: Initial-placement method comparison. (Left) Total expected production vs expected-production Gini. (Right) Distribution of mean entropy across methods. QP provides the strongest diversity signal, while GREEDY maximizes production.

To further diagnose inequality, Figure 4 reports the max/min ratio of expected production and the distribution of production gaps from the strongest player. Both QP and GREEDY compress extreme disparities relative to random baselines.

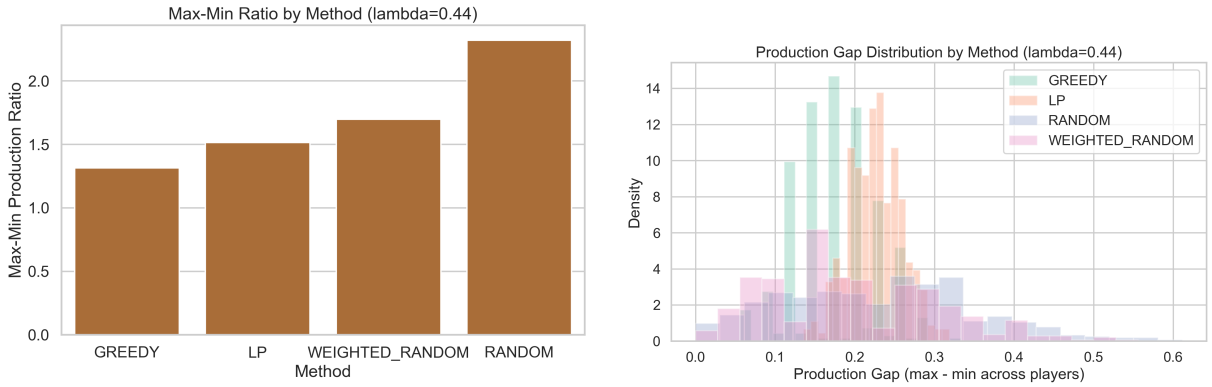


Figure 4: Initial-placement inequality diagnostics. (Left) Max/min ratio of expected production by method. (Right) Histogram of production gaps from the strongest player.

4.1.3 Full-Game Evaluation: Competitiveness and Outcome Parity

We now evaluate end-to-end gameplay across all 35 three-agent lineups. Figure 5 summarizes competitive performance. AB and VALUE dominate win rate and final VP, while CONVEX is not competitive on these metrics. This is expected, since CONVEX is designed to regulate equity rather than directly optimize winning.

Table 3 quantifies this tradeoff, reporting win rate and final VP (with confidence intervals), together with VP Gini, average game length, and average decision latency.

Agent	Win Rate (95% CI)	Final VP (95% CI)	VP Gini	Turns	Average Action-Decision Time (ms)
AB	0.858 [0.850,0.865]	9.45 [9.42,9.49]	0.309	79.5	11.53
VALUE	0.798 [0.790,0.807]	9.17 [9.13,9.20]	0.315	81.4	0.70
MCTS	0.203 [0.194,0.212]	6.13 [6.07,6.18]	0.281	164.3	22.66
WR	0.187 [0.178,0.196]	6.00 [5.95,6.06]	0.279	159.3	0.01
R	0.117 [0.109,0.125]	5.46 [5.40,5.51]	0.271	167.4	0.00
GREEDY	0.109 [0.101,0.117]	5.54 [5.48,5.59]	0.276	169.6	0.00
CONVEX	0.053 [0.048,0.058]	3.79 [3.75,3.83]	0.292	172.6	3.89

Table 3: Per-agent summary aggregated across all 35 lineups (500 games each).

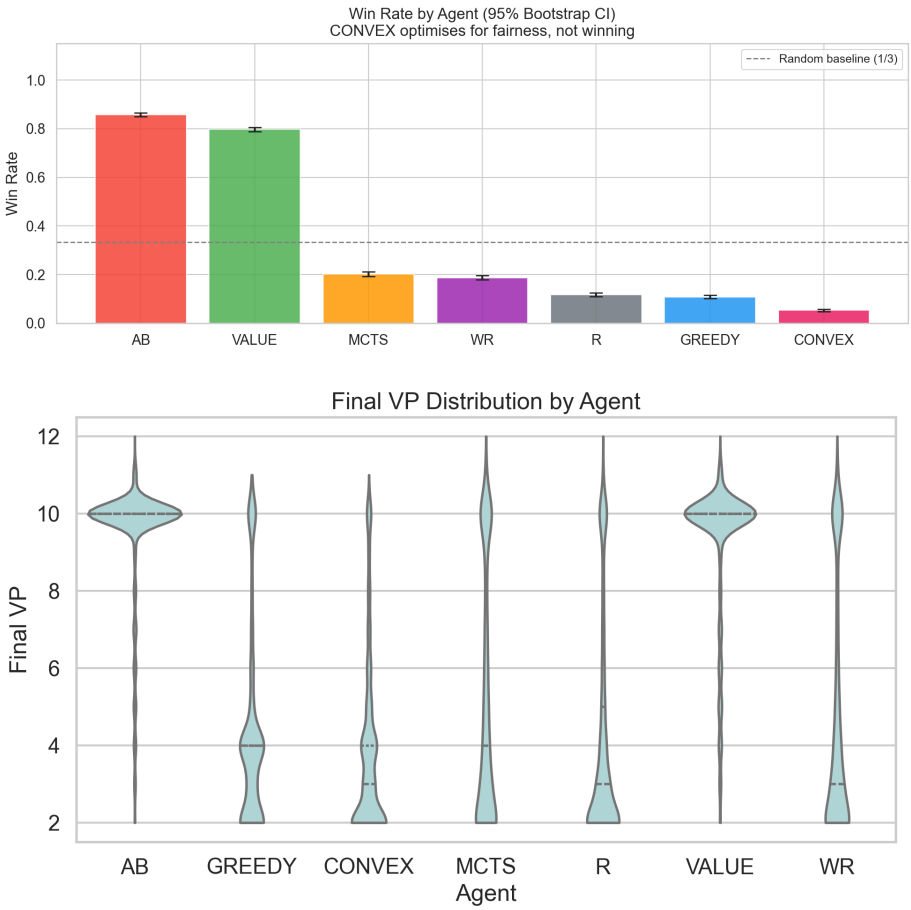


Figure 5: Full-game results across 35 lineups (500 games each). (Left) Win rate with confidence intervals by agent. (Right) Distribution of final victory points by agent.

The same trend is visible numerically in Table 3, which shows that stronger search-based agents (AB, MCTS) incur higher per-move compute, while lightweight baselines (R, WR, GREEDY) act nearly instantly.

Fairness in full games is evaluated using outcome parity metrics rather than expected-production metrics. Figure 6 shows VP Gini by lineup, along with a fairness-oriented summary plot (Gini-based) produced by our analysis scripts. Across all lineups, those containing CONVEX exhibit a small shift toward lower VP Gini on average, but the effect is modest compared to the variability induced by opponent strength.

Our primary fairness result is that incorporating CONVEX reduces outcome inequality as measured by the VP Gini coefficient. Across the set of evaluated lineups, lineups that include CONVEX exhibit a lower average VP Gini than lineups without CONVEX, indicating that the fairness objective generalizes across diverse opponents rather than overfitting to a single matchup. This supports CONVEX as a robust fairness-oriented allocator, even though it does not maximize win rate.

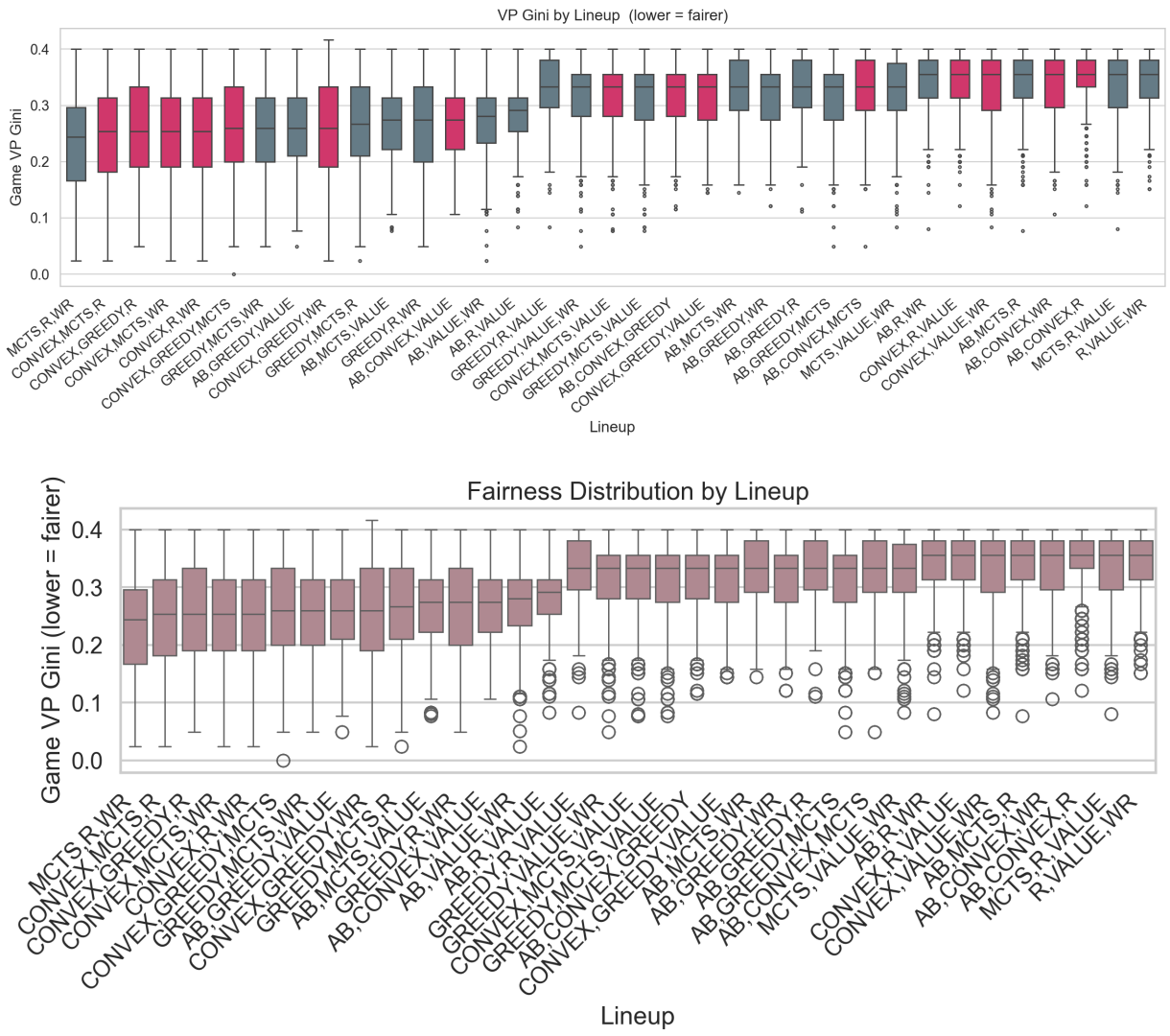


Figure 6: Outcome parity (VP inequality). VP Gini by lineup. Lower values indicate more equal end-of-game outcomes.

To make the lineup-level parity differences easier to interpret, Table 4 lists the five most equal and five most unequal lineups by VP Gini. A notable pattern is that the lowest-inequality lineups correspond to substantially longer games, suggesting that outcome parity can arise either from fairness regulation or from uniformly weak competitiveness that slows the entire game.

Lowest VP Inequality Lineups		
Lineup	**VP Gini**	**Turns**
MCTS,R,WR	0.233	278.8
CONVEX,MCTS,WR	0.244	274.0
CONVEX,MCTS,R	0.244	318.9
CONVEX,R,WR	0.246	295.1
CONVEX,GREEDY,R	0.249	343.7

Highest VP Inequality Lineups		
Lineup	**VP Gini**	**Turns**
AB,CONVEX,R	0.343	78.8
AB,R,WR	0.341	78.0
R,VALUE,WR	0.339	78.4
AB,MCTS,R	0.339	76.7
CONVEX,R,VALUE	0.339	80.5

Table 4: Lineups with the lowest and highest average VP Gini (outcome inequality) in the full-game evaluation. Very low-Gini lineups tend to correspond to much longer games.

We emphasize that the entropy metric here is computed over *end-of-game resource holdings* (the five resource counts in hand), not expected production. Figure 7 measures Shannon entropy of end-of-game resources held. This highlights differences in how agents tend to accumulate and retain diverse resources by the end of the game. Entropy is not perfectly aligned with competitiveness: some high-performing agents may still maintain diverse holdings, and some fairness-oriented behaviors may not maximize entropy in this particular definition.

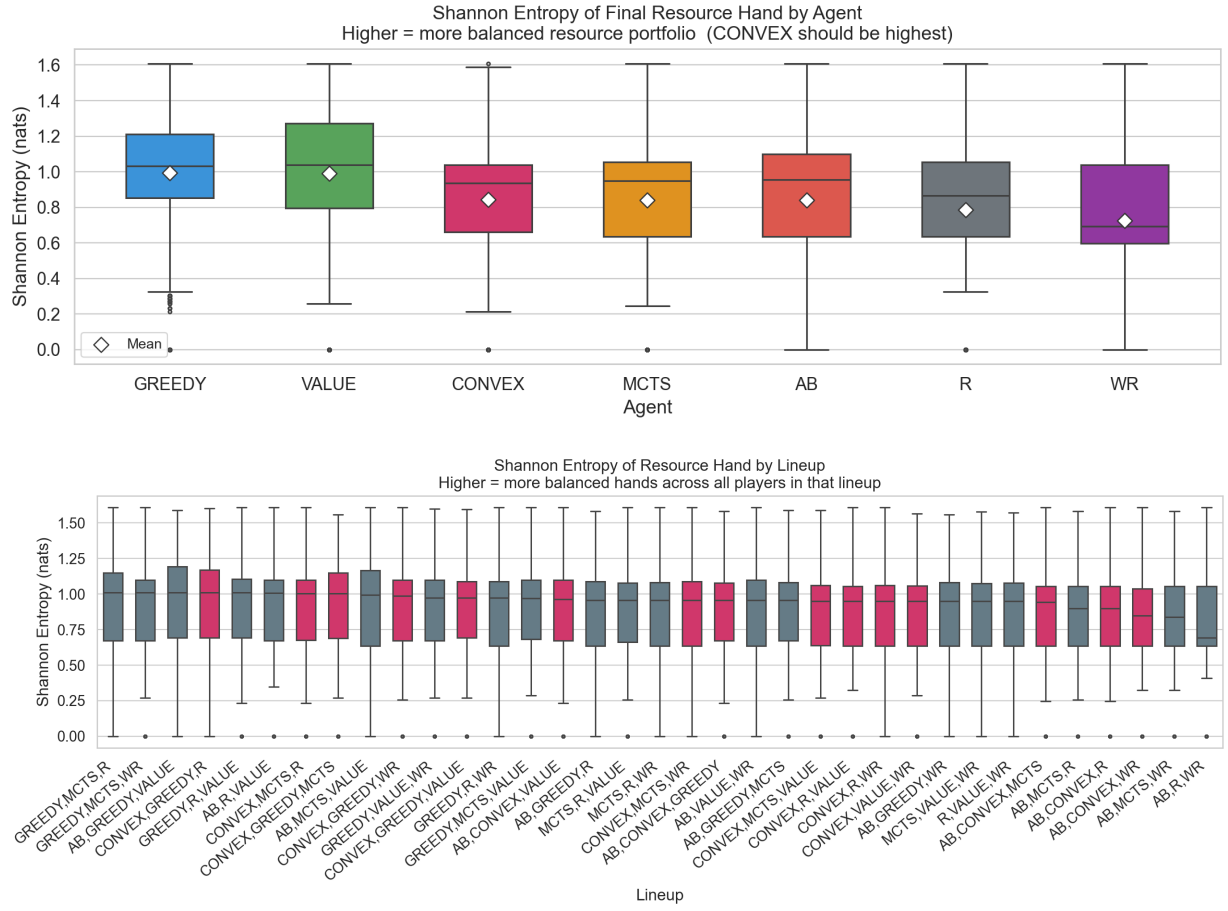


Figure 7: Resource diversity (end-of-game holdings). Shannon entropy computed from resources in hand. We report both per-agent and per-lineup views.

To understand matchups directly, Figure 8 shows the pairwise VP differential heatmap. This provides a

compact view of dominance relationships among agents and reinforces that strong adversaries (AB, VALUE) maintain a large advantage against all alternatives, including CONVEX. The result motivates future work that integrates fairness regulation with explicitly competitive objectives.

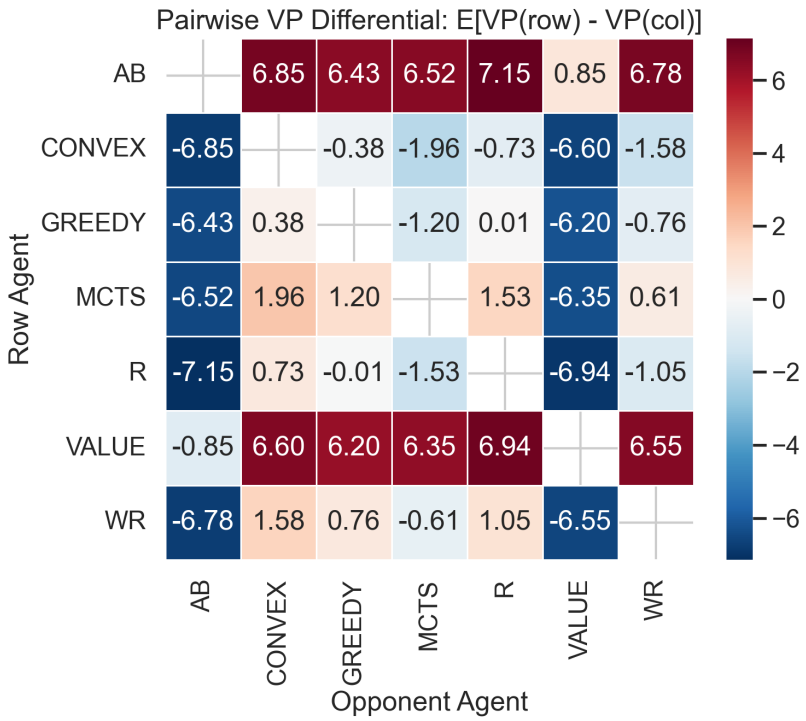


Figure 8: Pairwise VP differential heatmap. Entry (i, j) summarizes how agent i performs against agent j in VP differential.

Finally, Figure 9 summarizes the tradeoffs among fairness, performance, and compute. This is useful for assessing whether a fairness-oriented agent is practical and for selecting regimes where fairness improvements are obtained without excessive latency.

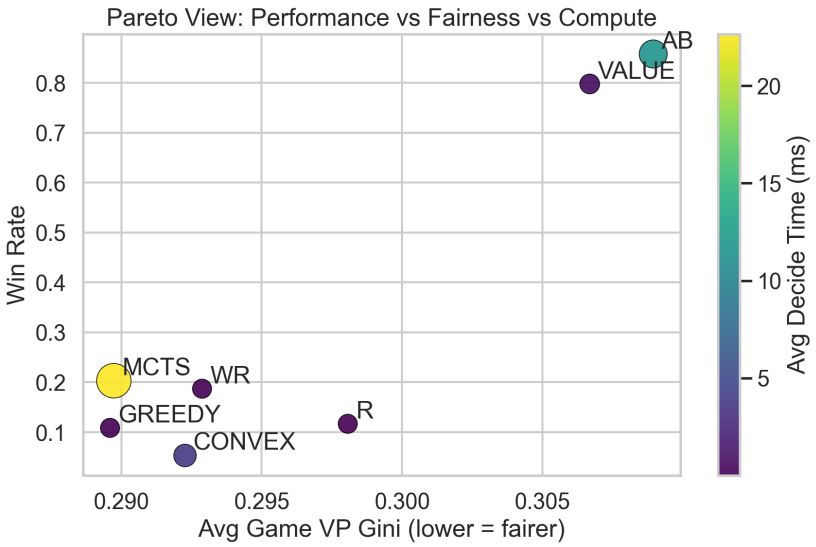


Figure 9: Pareto tradeoffs: fairness, win rate, and compute. We compare agents along axes capturing competitive performance, fairness-style inequality, and decision-time cost.

4.1.4 Turn Order Effects

Seat	Win Rate	Final VP	Avg. Rank
Seat 1	0.339	5.76	1.915
Seat 2	0.327	5.73	1.918
Seat 3	0.330	5.75	1.915

Table 5: Overall seat effects aggregated across all agents and lineups (weighted by number of seat appearances).

As shown in Table 5, aggregate seat effects are relatively small in our runs, which helps isolate agent and lineup effects as the primary drivers of the differences in Figures 5–9.

Turn order can systematically advantage earlier placements in Catan. Figure 10 visualizes turn-order effects in both VP outcomes and win rate. These plots help disentangle whether observed inequality is driven by the agent design versus structural seat advantage.

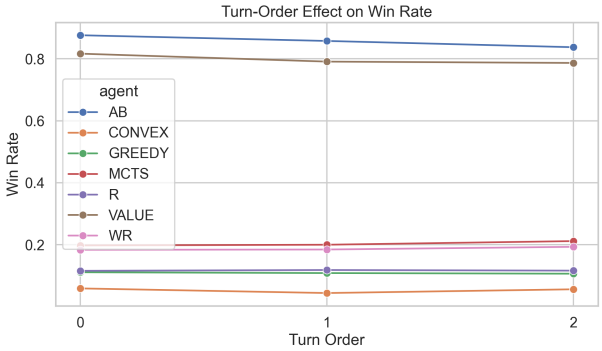


Figure 10: Effects of turn order. Outcome variation by seat provides a control for structural advantages unrelated to agent policy.

4.2 Conclusions

We draw the following conclusions from the QP-only and full-game experiments:

- The Convex Agent reduces inequality across diverse opponents (Table 4, Figure 6).
  - Across all evaluated lineups, introducing CONVEX lowers the VP Gini coefficient on average, showing it acts as a robust fairness-oriented resource allocator rather than a strategy tuned to a specific opponent.
- In the QP-only initial placement setting, λ is not the dominant lever over the tested range (Figure 2).
  - For QP, expected-production Gini, total expected production, and entropy vary only minimally over $\lambda \in [0, 2]$ (Figure 2).
- Method choice matters substantially in initial placement (Table 2, Figure 3).
  - Greedy maximizes total expected production, while QP yields substantially higher diversity (entropy) at a moderate efficiency cost. Random baselines are consistently worse in both fairness and efficiency (Table 2).
- In full games, competitive agents dominate (Table 3, Figure 5).
  - AB and VALUE achieve the highest win rates and final VP, indicating that fairness regulation alone is insufficient to compete against strong adversaries (Table 3).
- Outcome-parity improvements from fairness regulation are modest in gameplay (Figure 6, Table 4).
  - VP inequality shifts slightly across lineups containing CONVEX, but opponent strength and seat effects remain major drivers of variability (Figure 6, Table 5).

4.3 Future Work

- **Hybrid fairness-competitive objectives.** Incorporate VP-gain or expansion-potential signals into the QP objective alongside Δ^{opp}, enabling a Pareto-optimal tradeoff between fairness and win rate.
- **Deeper lookahead.** Extend the single-turn QP horizon to a multi-step MPC-style formulation, or integrate the fairness QP as a heuristic within fairness-regularised MCTS to remain competitive against strong agents.
- **Dynamic parameter scheduling.** Learn or schedule λ, μ, and ϕ as functions of game state (turn number, VP gap, resource counts) rather than fixing them as constants.
- **Non-maritime trading.** Model player-to-player trade proposals as a separate convex sub-problem, enabling Convex to offer resource trades that directly benefit the weakest opponent.
- **Port acquisition strategy.** Add port targeting to the QP: nodes adjacent to $2:1$ ports should receive a bonus proportional to the resources Convex is most likely to overproduce, enabling more efficient maritime trades.

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A Notation

- P: number of players; $p \in \{1, \ldots, P\}$; c denotes CONVEX.
- $L = 54$: settlement nodes; $l \in \{1, \ldots, L\}$.
- $K = 5$: resource types (wood, brick, sheep, wheat, ore); $k \in \{1, \ldots, K\}$.
- $A \in \mathbb{R}^{L \times K}$: production matrix; A_{lk} = expected per-turn production of resource k at node l.
- $\mathbf{w} = \mathbf{1} \in \mathbb{R}^K$: uniform resource weight vector.
- $x_{pl} \in [0, 1]$: relaxed placement variable ($\{0, 1\}$ in the discrete game).
- $X \in \mathbb{R}^{P \times L}$: placement matrix; row $\mathbf{x}_p$ for player p.
- $\boldsymbol{\rho}_p = \mathbf{x}_p A \in \mathbb{R}^K$: expected resource production vector of player p.
- $v_p = \boldsymbol{\rho}_p \cdot \mathbf{w}$: scalar production score.
- $\mathbf{v} = (v_1, \ldots, v_P)$: vector of production scores.
- b_p: placement budget; initial $b_p = 2$; mid-game $b_p = \max(0, 5 - |\mathcal{S}_p| - |\mathcal{C}_p|)$.
- $\mathcal{S}_p, \mathcal{C}_p$: sets of nodes with settlements and cities for player p.
- $\mathcal{O}$: set of currently occupied nodes.
- $\mathcal{E}$: set of board edges (adjacent node pairs); $|\mathcal{E}| = 72$.
- $\mathcal{H}(l)$: set of hexes adjacent to node l.
- n_h: number token of hex h; $p_{n_h} = (6 - |n_h - 7|)/36$.
- t^{opp}: auxiliary variable lower-bounding minimum opponent production.
- $v^{\text{opp}}_{\max}$: auxiliary variable upper-bounding maximum opponent production.
- $\Delta^{\text{opp}} = v^{\text{opp}}_{\max} - t^{\text{opp}}$: opponent production spread.
- $\lambda \geq 0$: compression weight (default 0.8).
- $\mu \geq 0$: self-balance penalty weight (default 0.05).
- ϕ: phase-adaptive floor fraction (equation (3)).

- $\bar{a} = \frac{1}{|\{l:\sum_k A_{lk} w_k > 0\}|} \sum_{l:\sum_k A_{lk} w_k > 0} \sum_k A_{lk} w_k$: board-average expected production per non-empty node.
- $\Phi_{\text{CONVEX}}(X)$: self-balance penalty (equation (2)).
- v_c^{cur}: current realized production of CONVEX from existing buildings.
- $\mathbf{q}_0 \in \mathbb{R}^K_{\geq 0}$: current resource hand.
- $\rho \in \{2, 3, 4\}$: best available maritime trade ratio.

B Dice Probability Model

Catan uses two six-sided dice. The probability that their sum equals s is $p_s = (6 - |s - 7|)/36$ for $s \in \{2, \ldots, 12\}$. Rolling a 7 activates the robber and yields no resources, so productive sums are restricted to $s \in \{2, 3, 4, 5, 6, 8, 9, 10, 11, 12\}$.

s	2	3	4	5	6	7	8	9	10	11	12
p_s	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

C Board Graph and Robber-Adjusted Matrix

The board has 19 resource hexes, $L = 54$ nodes, and $|\mathcal{E}| = 72$ edges. Each node is adjacent to up to three hexes. When the robber occupies hex h^*, all entries for adjacent nodes are zeroed:

$$\tilde{A}_{lk} = \begin{cases} 0 & l \text{ adjacent to } h^*, \\ A_{lk} & \text{otherwise.} \end{cases}$$

The matrix $\tilde{A}$ replaces A in all QP solves during mid-game turns.

D Full KKT Conditions

Using the Lagrangian multipliers, the KKT conditions are as follows.

Primal feasibility. Constraints (5)–(12) must hold.

Dual feasibility. All multipliers non-negative (except ν_l^p unrestricted).

Stationarity. Equations (13) and (14) for all p, l.

Complementary slackness.

$$\alpha_p(b_p - \textstyle\sum_l x_{pl}) = 0 \;\forall p, \tag{20}$$
$$\beta_l\Big(1 - \textstyle\sum_p x_{pl}\Big) = 0 \;\forall l, \tag{21}$$
$$\sigma_{lm}\Big(1 - \textstyle\sum_p x_{pl} - \sum_p x_{pm}\Big) = 0 \;\forall (l, m) \in \mathcal{E}, \tag{22}$$
$$\gamma_p(v_p - t^{\text{opp}}) = 0 \;\forall p \neq c, \tag{23}$$
$$\delta_p(v_{\max}^{\text{opp}} - v_p) = 0 \;\forall p \neq c, \tag{24}$$
$$\eta(v_c - \phi \bar{a} b_c) = 0, \tag{25}$$
$$\mu_{pl}^{+}(1 - x_{pl}) = 0 \;\forall p, l, \tag{26}$$
$$\mu_{pl}^{-} x_{pl} = 0 \;\forall p, l. \tag{27}$$

The KKT conditions are necessary and sufficient for global optimality because the primal is a convex QP with strong duality.

E Fallback Dual-Criterion Scoring

When all solvers fail, node l is scored by

$$s_l = 0.6\, s_l^{\text{block}} + 0.4\, s_l^{\text{help}},$$

where, letting $\delta_{\text{weak}} = \max(0, v_{\text{med}}^{\text{opp}} - v_{\text{min}}^{\text{opp}})$:

$$s_l^{\text{block}} = (\textstyle\sum_k A_{lk} w_k) \cdot \begin{cases} 1.5 & l \text{ adjacent to leader's nodes,} \\ 1.0 & \text{otherwise,} \end{cases}$$

$$s_l^{\text{help}} = (\textstyle\sum_k A_{lk} w_k) \cdot \begin{cases} 1.5 & l \text{ adjacent to weakest's nodes,} \\ 1.0 & \text{otherwise,} \end{cases} \cdot (1 + \delta_{\text{weak}}).$$

F Road-Unlock Heuristic

A road between nodes a and b is scored by the maximum fairness score of unoccupied nodes reachable from either endpoint:

$$s_{(a,b)}^{\text{road}} = \max_{l \in \mathcal{R}(a,b) \setminus \mathcal{O}} s_l,$$

where $\mathcal{R}(a,b)$ is the set of nodes adjacent to a or b. This captures the option value of future settlement placements.

G City Upgrade Scoring

A city adds $+1\times$ production at an already-occupied node (settlement $1\times \to$ city $2\times$). The urgency-weighted score is

$$s_l^{\text{city}} = (\textstyle\sum_k A_{lk} w_k) \left(1 + \dfrac{\max(0,\, v_{\text{med}}^{\text{opp}} - v_c^{\text{cur}})}{v_{\text{med}}^{\text{opp}} + \varepsilon}\right),$$

where $v_c^{\text{cur}} = \sum_{l' \in \mathcal{S}_c} \sum_k A_{l'k} w_k + 2 \sum_{l' \in \mathcal{C}_c} \sum_k A_{l'k} w_k$.

H Trade QP

The full trade QP is:

$$\begin{aligned} \text{maximize}_{\mathbf{t}^{\text{in}}, \mathbf{t}^{\text{out}} \geq \mathbf{0}} \quad & f(\mathbf{q}) := \mathbf{1}^\top \mathbf{q} - \frac{1}{2} \left\| \mathbf{q} - \frac{\mathbf{1}^\top \mathbf{q}}{K} \mathbf{1} \right\|_2^2, \\ \text{where} \quad & \mathbf{q} = \mathbf{q}(\mathbf{t}^{\text{in}}, \mathbf{t}^{\text{out}}) \end{aligned}$$

subject to $\mathbf{q} = \mathbf{q}_0 + \mathbf{t}^{\text{in}} - \mathbf{t}^{\text{out}} \geq \mathbf{0}$ and $\mathbf{1}^\top \mathbf{t}^{\text{in}} \leq \rho^{-1} \mathbf{1}^\top \mathbf{t}^{\text{out}}$. A trade is executed only if $\max_k t_k^{\text{out}} > 0.1$; give and receive resources are determined by $\arg\max \mathbf{t}^{\text{out}}$ and $\arg\max \mathbf{t}^{\text{in}}$.

I Robber Placement

$$h^* = \arg\max_h \; p_{n_h} \sum_{l \in \mathcal{N}(h)} \Big(\mathbb{1}[l \in \mathcal{S}_{\text{leader}}] + 2\, \mathbb{1}[l \in \mathcal{C}_{\text{leader}}] \Big), \tag{28}$$

where $\mathcal{N}(h)$ are nodes adjacent to hex h and the leader is $\arg\max_p v_p^{\text{cur}}$.

J Entropy-Based Discard

Shannon entropy of resource hand $\mathbf{q}$:

$$H(\mathbf{q}) = -\sum_k \frac{q_k}{\|\mathbf{q}\|_1} \log \frac{q_k}{\|\mathbf{q}\|_1}.$$

The agent discards

$$r^* = \arg\max_r H(\mathbf{q}_0 - \mathbf{e}_r), \tag{29}$$

keeping the remaining hand as balanced as possible.

K Self-Balance Regularisation

The penalty Φ_{Convex} (2) is zero if and only if $\boldsymbol{\rho}_c = \bar{\rho}_c \mathbf{1}$, i.e., equal expected production across all resource types. Its gradient with respect to x_{cl} is

$$\frac{\partial \Phi_c}{\partial x_{cl}} = 2 \sum_k \big(\boldsymbol{\rho}_{c,k} - \bar{\rho}_c\big)\big(A_{lk} - \tfrac{1}{K}\textstyle\sum_{k'} A_{lk'}\big)\,.$$